

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Neuromorphic DVS Events (REACT, ES-Parkour)
- TS + Proprioception / Telemetry (ACT, HiPolicy)
- TS + Quantum Circuits (Quantum-Mamba)
- TS + Causal Hypergraphs (CSHT, TiMi)

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
- Spiking State Space Scan (REACT, SpikySpace)
- Conditional Score Diffusion (CSDI, FADTI)
- Action Chunking & Diffusion (ACT, HiPolicy)
- Riemannian Hypergraph Trans. (CSHT)
- Continuous Neural ODE / CDE (LLMODE, SOTER)

3. Non-TS Role

- Auxiliary Context / Condition
- Microsecond Event Stream Trigger
 - Fourier Spectral Prior (FADTI)
- Macro Causal Shock Prior (CSHT)
- Embodied Visuomotor Goal Anchor
- Formal Invariant Certificate (SELA)

4. Downstream Tasks

- Multimodal Forecasting (Point / Conformal)
- Extreme Sensor Burst Imputation (CSDI, FADTI)
- High-Speed Agile Robot Parkour & Avoidance
- Financial Regime Shock Prediction (CSHT)
- Embodied Robotic Manipulation & Control
 - Safety-Critical CPS Logic Verification

Application Domains: Robotics & Manipulation • Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Industrial IoT & 6G